7. Domain: Communication Technologies

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	**PEARS (Programmable platform for Experiments and Academic Research on SDR)** is an indigenous SDR platform, offering a compact and cost-effective solution for research community to actively engage in hands-on exploration of wireless communication. This platform facilitates experimentation with a wide range of wireless technologies, from legacy AM/FM communications to state of the art technologies such as 4G-LTE and cognitive radio. PEARS also serves as an ideal platform for the swift and seamless implementation and validation of new ideas, algorithms, techniques and architectures.
	Name of the Project under which technology is developed	Development of SDR Development Platform for Academia & Research
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	8/9
	Brief Write-up on the Technology Developed	PEARS (Programmable platform for Experiments and Academic Research on SDR) is an indigenous SDR platform, offering a compact and cost-effective solution for research community to actively engage in hands-on exploration of wireless communication. This platform facilitates experimentation with a wide range of wireless technologies, from legacy AM/FM communications to state of the art technologies such as 4G-LTE and cognitive radio. PEARS also serves as an ideal platform for the swift and seamless implementation and validation of new ideas, algorithms, techniques and architectures. **Features** • Frequency coverage from 30 MHz to 2.4 GHz • Two transmit chains and two receive chains • Up to 56 MHz of Rx instantaneous bandwidth • Supports sampling rate up to 61.44 MS/s • Dual core ARM processing system • FPGA fabric for accelerated signal processing • Excellent signal processing power with 400+ DSP slices • Comprehensive support for GNU Radio environment **Typical use cases** • Rapid prototyping of wireless systems/solutions • Verification and validation of new algorithms, methods and designs • Develop custom/proprietary wireless protocols • Analysis and investigation of RF protocols • Training/Handholding for students and professionals in wireless
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Rakesh M, C-DAC Thiruvananthapuram rakeshm@cdac.in 9946772570

Sr. No.	Parameters	Brief details
2.	**Brief details of Technology Developed**	**TErrestrial Trunked RAdio (TETRA)** is a digital trunked mobile radio standard developed by the European Telecommunication Standards Institute (ETSI) to meet the needs of Professional Mobile Radio (PMR) user organizations. It offers clear, secure and reliable communication making it ideal for public safety, emergency services and mission-critical applications. The strong security features offered by TETRA standard makes it a good choice in strategic and military communications as well.
	Name of the Project under which technology is developed	TETRA System for Critical Communication
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	7/8
	Brief Write-up on the Technology Developed	**C-DAC TETRA Network (CTN)** is an indigenously designed wireless communication solution based on the ETSI TETRA Standard (ETSI 300 392-2) and offering unmatched reliability, seamless interoperability, advanced security and intuitive usability. It is a robust radio network having a scalable architecture allowing economic network deployments based on customer needs. **Features** • Guaranteed availability • Fast call set up (setup time of the order of 1/3rd of a second) • Individual, group & broadcast calls and messages • Simplex (PTT) and Duplex calls • Call pre-emption, priority calls, emergency calls • Supplementary services like Late Entry, Call Authorized by Dispatcher, Ambience Listening, Discreet Listening, Dynamic Group Number Assignment (DGNA) etc. • Authentication & Encryption • Adaptable soft-switching technology coupled with distributed database architecture. • Simple IP connectivity between components • Network resilience **Application Areas** ▪ *Public safety (Police, Fire, Ambulance etc.)* ▪ *Disaster management* ▪ *Military/Paramilitary* ▪ *On-board communication system for Naval platforms* ▪ *Railway signaling and communication, including Metro rails* ▪ *Seaports and Airports* ▪ *Industrial communication and so on*
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. John Thomas, C-DAC Thiruvananthapuram jontoms@cdac.in 0471-2723333 x. 3292

Sr. No.	Parameters	Brief details
3.	**Brief details of Technology Developed**	NLoS-Vicinity is a cutting-edge indoor positioning system that leverages Visible Light Communication (VLC) and Non-Line-of-Sight (NLoS) technology to deliver highly accurate user localization. Designed with user comfort in mind, it offers a novel solution ideal for smart proximity advertising and advanced location-based services.
	Name of the Project under which technology is developed	Visible Light Communication based LED Lighting Solutions
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	**Key Features** • Non-Line of Sight (NLoS) • Smartphone based user positioning • User Friendly **Key Use case** • Smart Proximity Advertising • Location based Services
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dhivya G, C-DAC Chennai dhivyag@cdac.in 9789944905
4.	**Brief details of Technology Developed**	VIsibLe Light CommUnication enabled SMart Indoor lightiNgAnd control SysTEm (ILLUMINATE) ILLUMINATE is a smart indoor lighting and control system that utilizes Visible Light Communication (VLC) technology. Tailored for RF-restricted environments like hospitals, sensitive industrial zones, and airports, it offers a reliable, interference-free solution for intelligent lighting and data communication.
	Name of the Project under which technology is developed	Visible Light Communication based LED Lighting Solutions
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 7
	Brief Write-up on the Technology Developed	**Key Features** • Quad-directional Full Duplex VLC Transceiver • Zone Control to avoid Visual Discomfort • Suitable for RF Free Zones • Distributed Control **End User** Smart lighting control in RF Free Zones
	Details of Contact Person alongwith Mobile No. and e-mail ID	Dhivya G, C-DAC Chennai dhivyag@cdac.in 9789944905

Sr. No.	Parameters	Brief details
5.	**Brief details of Technology Developed**	**C-DAC Radio Monitoring System (RMS)** is designed for wideband spectrum monitoring and offers a wide instantaneous digitization bandwidth. The product utilizes software-defined architecture for flexible feature addition and performance enhancements via waveform upgrades. It includes a Graphical User Interface (GUI) for platform configuration, data visualization, data capture and remote access to the RMS, all accessible through a desktop/laptop PC connected via Ethernet interface.
	Name of the Project under which technology is developed	Development of Radio Monitoring System
	Funding Agency	Internal Funding
	Technology Readiness Level (TRL)	5/6
	Brief Write-up on the Technology Developed	C-DAC Radio Monitoring System (RMS) is designed for wideband spectrum monitoring and offers a wide instantaneous digitization bandwidth. The product utilizes software-defined architecture for flexible feature addition and performance enhancements via waveform upgrades. It includes a Graphical User Interface (GUI) for platform configuration, data visualization, data capture and remote access to the RMS, all accessible through a desktop/laptop PC connected via Ethernet interface. **Major Features** • SDR based receiver architecture • Frequency coverage: 100 MHz – 2 GHz • Instantaneous bandwidth up to 192 MHz • Sweep rate of the order of 192 GHz/s at resolution bandwidth of 100 kHz • Noise Figure 11 dB (maximum) • Third Order Intercept +5 dBm @ input • Phase Noise -115 dBc/Hz @ 20 kHz offset • Configurable resolution bandwidth • Feature extraction capability • On-board hardware acceleration support for real-time analysis, demodulation • IQ data capture for offline analysis • Data visualization: Spectrum, spectrogram **Typical use cases** • Signal Intelligence • Transmission compliance • Interference management **User Segment** • Defense forces • Border & coastal security agencies • Law enforcement agencies • Regulatory agencies • Wireless service providers • Equipment operators • Broadcasters • Facility managers

Sr. No.	Parameters	Brief details
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Shine K. P., C-DAC Thiruvananthapuram shine@cdac.in 9895170335